LECTURE 4b / Fall 2026

Equity-Price Lattices Under Real-World and Risk-Neutral Measures

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LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

- **Build a binomial lattice.** Derive node prices and node probabilities from the up factor, down factor, horizon, and transition probability.
- **Separate probability measures.** Explain why real-world probabilities support forecasting while risk-neutral probabilities support no-arbitrage valuation.
- **Test model purpose.** Evaluate probability of profit under the real-world measure and identify stylized facts excluded by a fixed-parameter lattice.

1 A Finite Map of Uncertain Prices

An equity price has many possible futures. A binomial lattice makes uncertainty computationally explicit by replacing that continuum with two moves per step. The model is deliberately simple: it gives a complete distribution, preserves positive prices when its move factors are positive, and supports backward valuation. Its simplicity also makes its omissions visible.

Let $S_0 > 0$ dollars per share be the initial ex-dividend price. Over each step of length $\Delta t > 0$ years, the price is multiplied by the constant up factor $u > 0$ or down factor $d > 0$, with $d < u$. Multiplication commutes, so a path with k up moves and $n - k$ down moves reaches the same node regardless of the order of those moves. This is the source of recombination (Fig. 1).

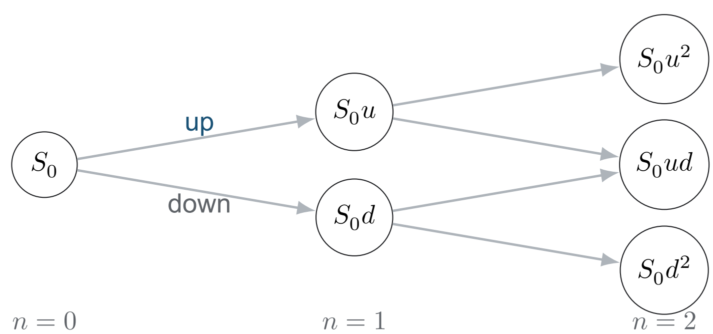

Figure 1: A two-step recombining equity lattice. The middle node is shared because $S_0ud = S_0du$. Recombination does not require $ud = 1$; that reciprocal constraint is an additional parameter choice used in some models.

2 Terminal Prices and Their Distribution

Assume first that successive moves are independent and that an up move has constant probability $p \in [0, 1]$. After n steps, the number of up moves K_n is binomial. This supplies both the node values and their probability mass (Prop. 2.1).

Proposition 2.1. Terminal distribution in an i.i.d. binomial lattice

Let $S_0 > 0$, $u > d > 0$, $p \in [0, 1]$, and let the positive integer n be the number of independent lattice steps. Let $K_n \sim \text{Binomial}(n, p)$ denote the number of up moves. Conditional on $K_n = k$, where $k \in \{0, \dots, n\}$, the terminal share price and its real-world probability are

$$S_{n,k} = S_0 u^k d^{n-k}, \qquad \mathbb{P}(K_n = k) = \binom{n}{k} p^k (1-p)^{n-k}. \tag{1}$$

The model has $n + 1$ distinct terminal nodes when $u \neq d$.

The expected gross return per step under this probability law is $pu + (1 - p)d$. Historical data can suggest estimates for u, d, and p, but the procedure is a model choice. Splitting past observations into positive and negative groups and averaging their move factors discards serial dependence, changing volatility, and tail structure. Estimated parameters also inherit sampling error and regime dependence.

COMPANION NOTEBOOK

L4b: Real-World Binomial Share-Price Simulation →

Estimate move factors and an up probability from historical data, populate a lattice, and compare its prediction bands with observed prices.

3 Two Measures, Two Questions

The symbol assigned to "probability of up" must identify its purpose. Under the real-world measure $\mathbb{P}$, p describes a statistical law for future prices and is estimated or forecast from data. Under a risk-neutral measure $\mathbb{Q}$, the up probability q is chosen so that discounted traded-asset prices satisfy the no-arbitrage pricing relation. It is generally not an empirical frequency.

Let the one-step risk-free gross return be $R_f > 0$. A risk-neutral lattice forces the expected share-price gross return under $\mathbb{Q}$ to equal R_f when dividends and frictions are absent. Solving that restriction gives Thm. 3.1.

Theorem 3.1. Risk-neutral probability in a one-step equity lattice

Let an equity with current price $S_0 > 0$ have next-step values $S_0 u$ and $S_0 d$, where $u > d > 0$. Let a locally risk-free asset have one-step gross return $R_f > 0$. If

$$d < R_f < u, \tag{2}$$

then the unique number $q \in (0, 1)$ satisfying the discounted-martingale condition is

$$S_0 = \frac{qS_0u + (1-q)S_0d}{R_f}, \qquad q = \frac{R_f - d}{u - d}. \tag{3}$$

Under a continuously compounded annual rate r_f over a step of Δt years, $R_f := \exp(r_f \Delta t)$. The result assumes no intermediate dividend and frictionless trading.

The inequalities in Eq. 2 are economic, not cosmetic. If $R_f \leq d$, the equity weakly dominates the risk-free asset in both states; if $R_f \geq u$, the risk-free asset weakly dominates the equity. Either boundary failure creates an arbitrage in the idealized one-period market or signals inconsistent inputs.

For any one-step payoff H_u dollars in the up state and H_d dollars in the down state, the same replication argument yields the no-arbitrage value

$$V_0 = \frac{qH_u + (1-q)H_d}{R_f}. \tag{4}$$

This is an expectation under $\mathbb{Q}$, not a statement that the real-world expected payoff equals the price grown at R_f. L10a will apply the recursion to options.

Table 1: The probability measure must match the question. Mixing columns produces incorrect forecasts or valuations.

	Real-world $\mathbb{P}$	**Risk-neutral** $\mathbb{Q}$
Primary use	Forecasting, scenarios, risk, probability of profit	No-arbitrage pricing and hedging
Up probability	Estimated or modeled from data	Implied by u, d, R_f in the complete one-step model
Expected equity return	Investor-required or data-generating return	Locally equals the risk-free return after discounting
Calibration test	Out-of-sample distribution and path behavior	Reproduce traded prices and exclude arbitrage

4 Probability of Profit Belongs to the Real-World Measure

Suppose a strategy costs $C_0 > 0$ dollars initially and has terminal liquidation value $W_{n,k}$ dollars at node (n, k). If profit means terminal wealth exceeding the initial cost grown at benchmark gross return R_b^n, the probability of profit is defined by Defn. 4.1.

Definition 4.1. Lattice probability of profit

Let $K_n \sim \text{Binomial}(n, p)$ under the real-world measure $\mathbb{P}$, let $W_{n,k}$ be terminal strategy wealth in dollars at node k, let $C_0 > 0$ dollars be initial cost, and let $R_b > 0$ be the benchmark gross return per step. The benchmark-relative probability of profit is

$$\text{POP}_{\mathbb{P}} := \mathbb{P}\Big(W_{n,K_n} > C_0 R_b^n\Big) = \sum_{k=0}^{n} \mathbf{1}_{\{W_{n,k} > C_0 R_b^n\}} \binom{n}{k} p^k (1-p)^{n-k}, \tag{5}$$

where $\mathbf{1}_A$ equals one when event A occurs and zero otherwise.

POP depends on the forecast law, horizon, benchmark, transaction costs, and profit definition. A large probability of a small gain can coexist with a small probability of a catastrophic loss, so POP is not expected profit and is never a sufficient risk measure. Using q from Eq. 3 in Eq. 5 would produce a pricing-measure probability, not the desired empirical success probability.

5 What the Fixed Lattice Cannot Reproduce

With constant u, d, and independent moves, one-step log returns take only two values and have zero serial correlation. Aggregated returns are affine transformations of a binomial count and approach a Gaussian shape after standardization as the step count grows. The model therefore captures weak autocorrelation by construction but not the heavy tails or volatility clustering documented in L4a. A prediction band can have nominal coverage in one sample while still failing exactly when regimes and volatility change.

This is not a reason to discard lattices. It is a reason to match the tool to the question. A lattice is excellent for exposing states, path-dependent trading rules, and replication. Forecasting rare equity moves requires richer innovations or state-dependent parameters, and every extension should be tested against data it was not fit to reproduce.

COMPANION NOTEBOOK

L4b: Lattice Models for Equity Prices →

Derive terminal node distributions and test which return stylized facts follow from a fixed-parameter binomial model.

KEY TAKEAWAYS

In this lecture, we built a recombining equity lattice, separated forecasting from pricing measures, and located probability of profit under the real-world law.

- **Recombination is algebraic.** A node depends on the count of up and down moves because multiplication commutes; the extra constraint $ud = 1$ is not required.
- **Measure follows purpose.** Real-world probabilities describe forecast frequencies, while risk-neutral probabilities enforce discounted no-arbitrage pricing.
- **POP is incomplete.** Probability of profit must use a stated real-world model and benchmark, and it does not reveal loss severity or expected profit.

Next, convert lattice states into explicit trading rules and quantify how entry, exit, and transaction-cost choices reshape probability of profit.

ADDITIONAL READING

- John C. Cox, Stephen A. Ross, and Mark Rubinstein, "Option Pricing: A Simplified Approach," *Journal of Financial Economics* 7(3), 229–263 (1979).
- John C. Hull, *Options, Futures, and Other Derivatives*, 11th ed. (2021), Chapters 13 and 15, for binomial trees, replication, and risk-neutral valuation.
- Rama Cont, "Empirical Properties of Asset Returns," *Quantitative Finance* 1(2), 223–236 (2001), for empirical validation targets omitted by the fixed lattice.